

Transition metals (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Identify d-block metals.
- Explain the high melting point and electrical conductivity of the transition elements.
- Compare d-block metals to s- and p-block metals.

So far in this subtopic about metals, we have largely ignored a large part of the periodic table containing metals that we often encounter in everyday life. When you think about examples of metals, what names come to mind? Some examples of metals that many are familiar with include iron as a metal used for car bodies, copper used in most electrical wiring and tungsten used to make extremely hard cutting and drilling tools. What region of the periodic table (**Figure 1**) are these elemental metals located?

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.01																	2 He 4.00
2	3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3	11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La † 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226)	89-103 Ac ‡ (227)	104 Rf (267)	105 Db (268)	106 Sg (269)	107 Bh (270)	108 Hs (269)	109 Mt (278)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (288)	116 Lv (293)	117 Ts (294)	118 Og (294)
†			58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97		
‡			90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)		

Figure 1. Periodic table.

 More information for figure 1

This image is a color-coded periodic table of elements. The table is divided into sections by colors, each representing different element groups. Elements are arranged in rows and columns. Each element's cell contains its atomic number, chemical symbol, and relative atomic mass. The color coding includes: alkali metals in orange, alkaline earth metals in yellow, transition metals in blue, lanthanides and actinides in red, metalloids in purple, nonmetals in pale green, halogens in green, and noble gases in violet. The table visually separates the lanthanides and actinides into a section below the main body of the table.

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Think back to when you first learned about electron configurations in [section S1.3.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>). What was it about the elements in groups 3–12 that were different to the elements in groups 1 and 2? Recall from [section S2.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>) that transition metals are capable of forming ions of different charges. For example, two common ions of copper include Cu^+ and Cu^{2+} . In [section S3.1.8](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) you learned that the transition metal ions form complex ions through covalent bonding of a ligand. Because these metals have valence electrons occupying d orbitals, many of their properties are unique in comparison to their s and p metal counterparts. In this section, we will look more closely at the metals located in this d block region of the periodic table and their associated properties.

Making connections

In [section S3.1.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/trends-in-atomic-and-ionic-radius-id-43570/>), periodic trends will be explored in more detail for s and p block elements. Why are periodic trends of the transition elements less evident? What is unique to the electronic components of these elements in comparison to the s and p block elements?

Recall from [section S1.3.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>) that transition elements have valence electrons occupying d orbitals. Metals in this region of the periodic table are consequently referred to as d block metals. **Figure 2** shows an example of a few of

these transition elements with their associated electron configuration. Note the terms 'transition elements' and 'transition metals' will be used interchangeably here as transition elements are metals.

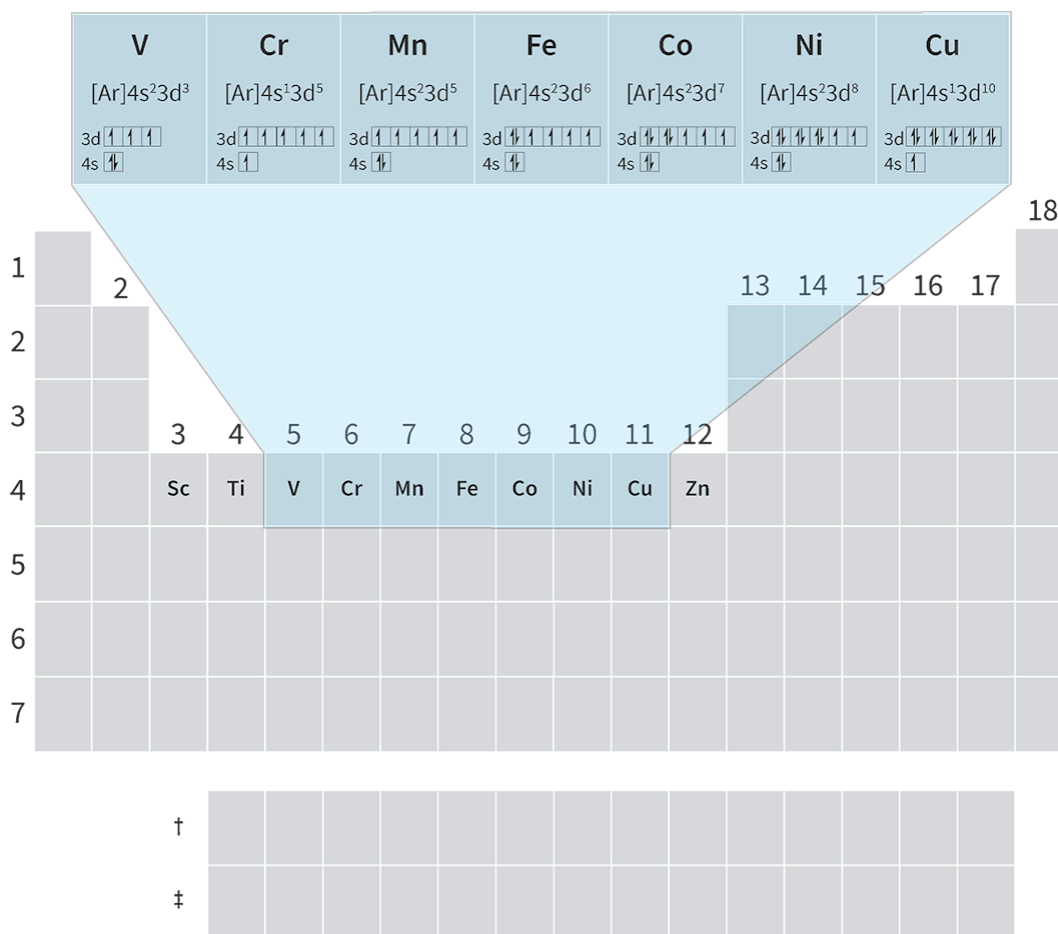


Figure 2. Some transition elements and their electron configurations.

More information for figure 2

This image is a section of the periodic table highlighting some transition elements along with their electron configurations. The section covers elements from group 5 to 12, which include V (Vanadium), Cr (Chromium), Mn (Manganese), Fe (Iron), Co (Cobalt), Ni (Nickel), and Cu (Copper). Above each element's symbol, their electron configuration is specified, showing the occupation of electrons in the d and s orbitals. For example, Vanadium (V) is shown to have the electron configuration [Ar] 4s² 3d³. Each element in this section is visually represented along with a pattern illustrating their electron distribution. The image highlights the placement and structure of these transition metals within the periodic table, providing a visual understanding of their electron configuration specifics.

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For transition metals, the valence electrons can be considered to occupy both the s and d orbitals. For example, transition elements located in period 4 have valence electrons occupying 4s and 3d orbitals. This means that there are many possible

ionic charges transition metals can form, as we saw in [section S2.1.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>). These electrons are loosely bound and delocalise throughout the metallic lattice.

As there are a large number of valence electrons from both the s and d orbitals, this results in a greater electron density within the metallic lattice. This increased electron density in turn increases the strength of the metallic bond, as we saw in [section S2.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/>), in comparison to s and p block metals.

Melting point

The melting points for period 4 metals are shown below in **Figure 3**. Notice how, in general, the melting points are much higher for the transition elements. This is due to the increased strength of the metallic bond. The large number of delocalised electrons from the 4s and 3d orbitals, increases the attraction they have with the metal ions in the lattice. Can you identify any metals in the d block that are not consistent with this trend of high melting point?

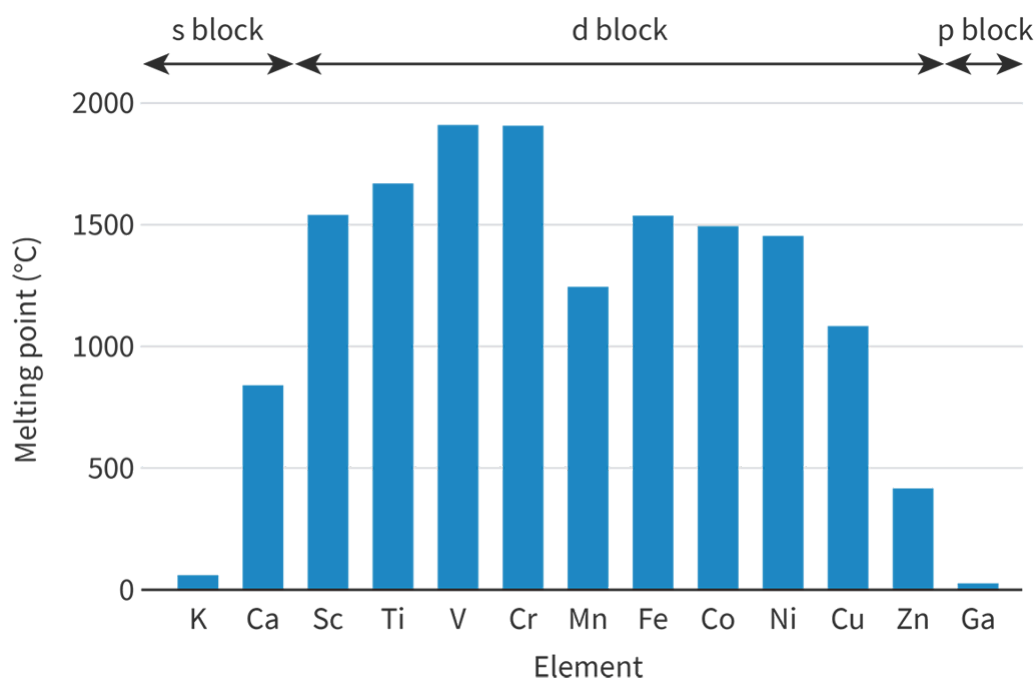


Figure 3. Melting points for period 4 metals.

 More information for figure 3

The image is a bar graph showing the melting points of period 4 metals. The x-axis is labeled with the names of the metals: potassium (K), calcium (Ca), scandium (Sc), titanium (Ti), vanadium (V), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), and zinc (Zn). The y-axis represents the melting point in degrees Celsius, ranging from 0 to 2000 degrees Celsius at intervals of 500. Tallest bars are seen for chromium (Cr) and vanadium (V) which have high melting points above 1500 degrees

Celsius. Notably, zinc (Zn) has a much lower melting point, depicted with a shorter bar. The graph highlights the trend where transition metals generally have higher melting points compared to other metals in period 4, except zinc.

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You might notice that zinc has a much lower melting point than the rest of the d block metals in period 4. In [section S3.1.8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-2-id-43595/>) you will learn more about transition elements. There you will learn about why zinc, although it is a d block metal, is not technically considered a transition element.

Theory of Knowledge

You have just learned the relationship between the melting point of a metal and electron density. The melting point across the period 4 metals has many inconsistencies in this trend. How do inconsistencies in trends reduce the reliability of relationships?

Hardness

You saw earlier in [section S2.3.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/>) how metals in group 1 were quite soft and could be easily sliced into. How would you expect transition elements to compare in hardness to metals in group 1? **Interactive 1** below shows someone cutting into potassium and copper, both period 4 metals. Are you able to identify the hard metal? Why do you think this is so?

Interactive 1. The Varying Hardness of Cutting Potassium and Copper.

 More information for interactive 1

Two videos titled Potassium and Copper. The video on the left, titled Potassium Metal is Like Weird Butter, focuses on potassium. The video demonstrates the physical properties of potassium metal. Potassium metal is taken out of a small tube and is then cut with a knife. The video shows that potassium is soft and can be easily cut like butter, revealing a shiny interior. The metal is pressed with fingers, revealing signs of crumbling and breaking. The cut piece of metal, when added to a container with water, reacts immediately emitting smoke and flame.

The video on the right for copper is titled Copper Sheet Thickness Guide 22 Mil (16 ounce copper). It provides a reference for the thickness of 22 mil copper sheeting, also known as 16 ounce copper. The video shows the copper sheet being folded easily. A scissor is used to cut the sheet and in to pieces. The video demonstrates the physical characteristics and flexibility of this specific copper thickness, offering users a better understanding of its properties. The video also directs users to a related thickness guide for more information.

Because copper has both 4s and 3d electrons as delocalised valence electrons, there is an increased attraction between these electrons and the metal ions in the lattice. This increased strength of metallic bonds results in a greater hardness of the copper in comparison to potassium.

Electrical conductivity

Transition elements are electrically conductive. These metals form their metallic bonds through the delocalisation of electrons in unfilled d orbitals. The electrostatic attraction between metal ions in the lattice and delocalised electrons increases with an increasing number of electrons in d orbitals. As there is free movement of electrons within these d orbitals, the transition metals are good conductors of electricity. Copper, for example, is one of the most conductive metals. For this reason, it is used in most electrical wiring as we saw in one of the examples earlier in **Figure 3**.

The properties discussed above are desirable properties for metals that allow it to be used in coins, jewellery, electrical wires, cookware, etc. **Interactive 2** shows common metal materials made from transition elements.



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Interactive 2. Common Metal Objects Made of Transition Metals.

Activity

- **IB learner profile attribute:** Inquirers
 - **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
 - **Time required to complete activity:** 30 minutes
 - **Activity type:** Pair activity
1. Create a Venn diagram comparing and contrasting the properties of transition metals vs. s block metals. Consider use of the data booklet or researching data to help support your entries into the Venn diagram.
 2. Design your own question to link this subtopic to another you have already learned about or to the nature of science.